

Exploring the Efficacy of Virtual Research Mentorship for Online Undergraduate Projects

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Abstract

Virtual undergraduate research mentorship is gaining traction, offering key support for student participation in high-impact research. However, limited research exists on its effectiveness. This study features an examination of faculty mentorship in online undergraduate research using the validated Mentoring Competency Assessment. While mentors rated themselves highly, mentees rated them even higher, suggesting effective virtual mentoring in combination with potential mentor self-confidence issues. These findings support virtual mentorship's value and highlight areas for targeted professional development.

Keywords: virtual mentorship, undergraduate research, mentor competencies

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Undergraduate research is a high-impact practice, defined by the Council on Undergraduate Research (CUR) as “a mentored investigation or creative inquiry conducted by undergraduates that seeks to make a scholarly or artistic contribution to knowledge” (Undergraduate Research Definition Task Force, n.d., para. 1). Mentorship is central to this experience, benefiting both students by improving research persistence (Cooper et al., 2019) and self-efficacy (Estrada et al., 2018; Sams et al., 2015) and faculty, who gain research assistance, mentoring skills, and student feedback (Faulconer et al., 2020a; King & Imai, 2022). However, mentoring presents challenges, including personal, student-centered, and institutional barriers (Faulconer et al., 2020a; King & Imai, 2022). Undergraduate curricula may lack research integration (Adebisi, 2022), and adjunct faculty, often students’ closest content experts, may have limited research experience. A central motivation to this research is to better understand how mentors evaluate their own competencies and how mentees perceive their guidance within a fully online, globally dispersed undergraduate research mentorship program. In this model, research mentors and supervisors fulfill distinct roles. The investigation seeks to answer: 1) *Do mentors self-perceive strong competencies across all six mentoring domains?*; 2) *Do mentees perceive strong competencies in their mentors?*; and 3) *Is there strong alignment between mentor self-assessments and mentee evaluations?*

Literature Review

Research often conflates the roles of a research mentor and supervisor, despite their distinct functions. While both share responsibilities, mentorship emphasizes long-term professional growth, guiding students from research curiosity to engagement, while supervision provides disciplinary expertise for project execution. The National Research Mentoring Network defines mentoring as a structured, evolving relationship focused on career progression (NRMN, n.d.), yet there remains a need for clearer definitions, role delineation, and relationship context (Pfund, 2016). Addressing faculty barriers to undergraduate research support could involve separating mentorship from supervision. Mentors would foster skill development, self-reflection, and career readiness, while supervisors focus on project-specific expertise. Virtual research mentorship is gaining attention, with models like mentor constellations and cohort mentoring showing promise (Hall et al., 2021; Johnson & Knox, 2022). Benefits include improved research understanding, employability, and inclusivity, especially for underrepresented groups (Faulconer et al., 2024; Knox et al., 2023), though direct comparisons between virtual and in-person mentoring remain scarce.

Studies highlight discrepancies between mentor and mentee perspectives on research skills and feedback. Mentees often underestimate their abilities, and mentors may lack sufficient interaction to assess skills accurately (Lev et al., 2010; Feldon et al., 2015). Differences in mentoring expectations also influence how feedback is received and applied (Korver & Tillema, 2014). Formal mentor training is critical, but underutilized, with only seven-percent reporting significant training (Stolzenberg et al., 2019). Effective in-person and online training can enhance cultural competence and mentoring effectiveness (Byars-Winston et al., 2020). Efforts to assess mentoring effectiveness have led to tools such as the Mentor Evaluation Tool (Yukawa et al., 2020) and the Mentoring Competency Assessment (Fleming et al., 2013), which aligns mentor and mentee perspectives. Notably, studies reveal gender disparities, with male mentors rating their skills higher than female mentees do, particularly in fostering independence and professional development (Orsini et al., 2019). Addressing these gaps through improved metrics and training can enhance mentoring relationships and research outcomes.

Methods

This study employed a structured methodology, detailing contextual roles, the mentoring program, materials, participant procedures, and the data analysis plan to ensure rigor and coherence. At the subject institution, research mentors and supervisors have distinct roles. Mentors provide long-term guidance, supporting students from initial research interest to project completion through psychosocial, role-modeling, and career functions (Abedin et al., 2012; Pfund, 2016; Thiry & Laursen, 2011). They meet regularly with mentees, often remotely, and coordinate with institutional mentoring programs. Supervisors, typically subject matter experts, focus on short-term project management, methodology, and data analysis, ensuring research completion. Coaching may fall under either role, with mentors emphasizing broader research concepts, such as funding and professional development. The voluntary research mentoring program matches students with mentors based on academic discipline and goals. Mentors, typically support 4–8 students, complete online training, and develop a mentoring philosophy. Participants come from diverse STEM fields, including physical sciences, digital humanities, and mathematics.

The Mentoring Competency Assessment (MCA), a validated tool with six subscales (communication, expectations, understanding, independence, diversity, and professional development) was used to evaluate mentor competencies (Fleming et al., 2013; Hyun et al., 2022). A 21-item survey utilized a 7-point Likert scale, with mentors self-assessing skills and mentees rating their mentors. Demographic data were also collected. Following Institutional Review Board approval (#23-103), participants were recruited via email and completed surveys through Qualtrics. The study included 29 respondents (23 mentees, six mentors), with an analyzed sample of eight (six mentees, two mentors). Response rates aligned with prior studies (Orsini et al., 2019; Wisker et al., 2022). Data Analysis focused on mentoring competency using t-tests to compare mentor and mentee ratings across six domains. Mentees were divided into Mentor Groups A and B, with Group C excluded due to mentor non-participation. Despite the small sample size (n=8), analysis provided preliminary insights to guide future research. Ratings of 6+ indicated strong competency, with the null hypothesis assuming an average rating of 6. Rejection of the null hypothesis confirmed perceived strong skills.

Results

This work represents an assessment of mentor and mentee competency perceptions, testing the null hypothesis that their average ratings were equal. Alignment was confirmed, if the null hypothesis was not rejected. Mentor self-assessments across six competencies showed no significant differences between Groups A and B, allowing their

ratings to be combined. Mentors rated themselves as strong (mean ≥ 6) in *Aligning Expectations* (6.63, $p \leq 0.05$) and in three competencies at the 0.10 level: *Aligning Expectations*, *Assessing Understanding* (6.33), and *Fostering Independence* (6.50). Mentees rated their mentors across the same competencies, with scores ranging from 6.44 (*Fostering Independence*, Group A) to 7.00 (*Addressing Diversity*, Groups A & B). No significant differences between mentee groups allowed for their ratings to be combined, leading to the null hypothesis being rejected at both significance levels; the result indicates mentees consistently viewed their mentors as highly skilled. Comparing mentor self-ratings to mentee evaluations, Group A showed alignment in four competencies, while Group B aligned in one (*Aligning Expectations*). In all cases of misalignment, mentees rated their mentors higher than the mentors rated themselves, suggesting mentors either concurred with mentee perceptions or underestimated their own competencies.

Discussion

The analysis of mentor and mentee perceptions identified areas of alignment and discrepancy in mentoring competencies. Mentors rated themselves highly in *Aligning Expectations*, *Assessing Understanding*, and *Fostering Independence* at the 0.10 significance level, with *Addressing Diversity* and *Promoting Professional Development* averaging 6 or higher. *Effective Communication* had the lowest average (5.75), yet mentees consistently rated their mentors higher across all competencies. While Mentor A's ratings aligned with mentees in four areas, Mentor B aligned in only one, suggesting differences in self-perception rather than skill deficits. *Effective communication* is crucial in mentoring (Bennett et al., 2022; VanAlstine & Holmes, 2018) yet can be challenging in virtual settings due to technological constraints (Owen, 2015). Mentees rated their mentors highly, though mentors saw room for improvement, mirroring in-person findings (Harker et al., 2019). Unlike in-person mentoring, where misalignment is common (Harker et al., 2019; Orsini & Stedman, 2022), virtual mentoring showed stronger alignment; particularly in *Assessing Understanding*, a critical factor in effective mentorship (Greenberg, 2018; Stephenson et al., 2022). However, misjudging mentee knowledge can hinder success, especially in cross-disciplinary mentoring, where targeted training may help. *Fostering Independence* is central to undergraduate research (Pfund, 2016; Walkington et al., 2020), with confidence playing a key role (Greenberg, 2018). While mentees rated this competency highly, one mentor rated themselves lower, aligning with prior research (Harker et al., 2019). *Effective mentorship* requires balancing guidance and autonomy through intentional strategies (Bennett et al., 2022; Ghebreyessus et al., 2022). *Diversity mentoring* also presents challenges as mentors influence career trajectories (Carpi et al., 2017; Jones & Lerner, 2019). Mentees rated diversity mentoring highly, though mentor self-assessments varied from past studies (Harker et al., 2019). *Faculty development* programs can enhance cultural competency (Byars-Winston et al., 2023), supporting diversity through networking and research dissemination (Walkington et al., 2020). Professional development ratings were high, but varied, which is consistent with in-person studies (Harker et al., 2019; Orsini & Stedman, 2022). Work-life balance (Bennett et al., 2022) and access to multiple mentors (Bradley, 2017) further contribute to effective mentoring.

Implications

This study highlights the strengths of virtual research mentoring, while identifying self-perception gaps between mentors and mentees. Mentees consistently rated mentors higher, underscoring the need for structured training to address perception discrepancies and improve diversity mentoring. Not all faculty excel in mentoring (Johnson et al., 2015), but online training enhances ability and confidence (Behar-Horenstein et al., 2019; Weber-Main et al., 2019), improving communication, expectation alignment, and engagement (Young & Stormes, 2020). Strong mentoring boosts research output and student belonging (Morales et al., 2017), with inexperienced students benefiting the most in online settings (Monarrez et al., 2020; Faulconer et al., 2020b).

Key Takeaways, and Future Direction

This work relied on refinement of Pfund's (2016) undergraduate mentoring framework to provide insights into virtual mentoring using a validated mentor-mentee comparison metric. While many institutions adopted virtual mentoring during COVID-19 (Erickson et al., 2022), long-term sustainability remains uncertain. Results exhibit mentor-mentee agreement in key competencies: communication, expectation alignment, assessing understanding, fostering independence, diversity, and professional development. Though a small sample limits generalizability, this study establishes a foundation for future research comparing virtual and in-person mentoring. For future efforts, a larger, more diverse sample would improve statistical reliability. Expanding the sample would allow degree-level comparisons, given that mentoring needs differ between undergraduates and graduate students (Gonzalez, 2001; Butz & Branchaw, 2020). Data from a single institution also limits generalizability, and a small mentor sample prevented analysis of heterogeneity effects. While gender was not a predictor of mentor self-evaluation, larger studies suggest that gender, research area, mentoring frequency, and academic rank may influence results (Mickel et al., 2018; Orsini et al., 2019). Bias is also a concern, as self-reported data may be influenced by social desirability, and voluntary participation could introduce selection bias. Nonresponse error may affect representativeness, though recruitment efforts aimed to mitigate this. Expanding participation would enhance reliability and applicability.

Conclusion

This study offers insights into the competencies of virtual research mentors, including the identification of strengths in areas such as communication, aligning expectations, and fostering independence. However, differences between mentor and mentee perspectives, particularly on communication, suggest areas for deeper exploration. The limited sample size restricts generalizability, indicating the need for larger, longitudinal studies to better assess the long-term impacts of virtual mentoring. Future research should also focus on how mentor development programs can enhance skills in key areas like diversity management and promoting mentee independence. Overall, this study sets the stage for future research into improving virtual mentoring practices and mentor training.

Declaration of Conflicting Interests

Authors disclose employment at the institution studied as a potential conflict of interest.

Acknowledgments

This work was supported by the National Science Foundation [grant numbers 2021221 and 2315560]. This paper represents a condensed version of a larger manuscript, titled “*Virtual Mentorship for Online Undergraduate Research: Analysis of Mentors and Mentees' Perspectives.*”

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